$$\frac{dx_1}{dt} = -x_1 + p_2(1+x_1)\, e^{\frac{x_2}{1+\frac{x_2}{20}}}$$

$$\frac{dx_2}{dt} = -x_2 + p_2 p_4 (1-x_1)\, e^{\frac{x_2}{1+\frac{x_2}{20}}} - p_5 x_2 =$$

$$= -(1+p_5)x_2 + p_2 p_4 (1-x_1)\, e^{\frac{x_2}{1+\frac{x_2}{20}}}$$

Пусть $E(x_2) = e^{\frac{x_2}{1+\frac{x_2}{20}}} = e^{\frac{20x_2}{20+x_2}}$

$$\frac{dx_1}{dt} = -x_1 + p_2(1-x_1)\, E(x_2)$$

$$\frac{dx_2}{dt} = -(1+p_5)x_2 + p_2 p_4 (1-x_1)\, E(x_2)$$

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix}$$

$$E'(x_2) = E(x_2)\cdot\left(\frac{20x_2}{20+x_2}\right)' = E(x_2)\,\frac{20(20+x_2)-20x_2}{(20+x_2)^2} =$$

$$= E(x_2)\,\frac{400}{(20+x_2)^2}$$

$$\frac{\partial f_1}{\partial x_1} = -1 - p_2 E(x_2) \qquad \frac{\partial f_1}{\partial x_2} = p_2(1-x_1)\,E(x_2)\,\frac{400}{(20+x_2)^2}$$

$$\frac{\partial f_2}{\partial x_2} = -p_2 p_4 E(x_2) \qquad \frac{\partial f_2}{\partial x_2} = -(1+p_5) + p_2 p_4 (1-x_1)\,E(x_2)\cdot \frac{400}{(20+x_2)^2}$$

$$J = \begin{bmatrix} -1 - p_2\, e^{\frac{20x_2}{20+x_2}} & p_2(1-x_1)\, e^{\frac{20x_2}{20+x_2}}\cdot\frac{400}{(20+x_2)^2} \\ -p_2 p_4\, e^{\frac{20x_2}{20+x_2}} & -(1+p_5) + p_2 p_4 (1-x_1)\, e^{\frac{20x_2}{20+x_2}}\,\frac{400}{(20+x_2)^2} \end{bmatrix}$$

Аутф.

$$-x_1 + p_2(1-x_1)\, e^{\frac{20x_2}{20+x_2}} = 0$$

$$-(1+p_5)x_2 + p_2 p_4 (1-x_1)\, e^{\frac{20x_2}{20+x_2}} = 0$$

Пусть $E(x_2) = e^{\frac{20x_2}{20+x_2}}$:

$$-x_1 + p_2(1-x_1)E(x_2) = 0 \quad (1) \qquad / \cdot p_4$$

$$-(1+p_5)x_2 + p_2 p_4 (1-x_1)E(x_2) = 0 \quad (2)$$

$(1)\cdot p_4$:

$$-p_4 x_1 + p_2 p_4 (1+x_1)E(x_2) = 0$$

$(2) + (1)\cdot p_4$:

$$\left(-(1+p_5)x_2 + p_2 p_4 (1-x_1)E(x_2)\right) - \left(-p_4 x_1 + p_2 p_4 (1+x_1)E(x_2)\right) = 0$$

$$-(1+p_5)x_2 + p_4 x_1 = 0$$

$$p_4 x_1 = (1+p_5)x_2$$

$$x_1 = \frac{(1+p_5)x_2}{p_4} \quad (4)$$

Выразим $p_2 E(x_2)$:

$$-x_1 + p_2(1-x_1)E(x_2) = 0$$

$$p_2(1-x_1)E(x_2) = x_1 \quad / : (1-x_1)$$

$$p_2 E(x_2) = \frac{x_1}{1-x_1} \quad (3)$$

Подставим x_1 в (3):

$$1 - x_1 = 1 - \frac{(1+p_5)x_2}{p_4} = \frac{p_4 - (1+p_5)x_2}{p_4}$$

$$p_2 E(x_2) = \frac{\frac{(1+p_5)x_2}{p_4}}{\frac{p_4-(1+p_5)x_2}{p_4}} = \frac{(1+p_5)x_2}{p_4-(1-p_5)x_2} \quad (5)$$

Подставим стационарные точки:

$$a_{11} + a_{22} = 0$$

$$a_{11} = -1 - p_2 E(X_2)$$

Подставим (5):

$$a_{11} = -1 - \frac{(1+p_5)X_2}{p_4 - (1+p_5)X_2} = -\frac{p_4}{p_4 - (1+p_5)X_2}$$

Из стац. ур-я: $X_1 = p_2(1-X_1)E(X_2)$

$$a_{12} = p_2(1-X_1)E(X_2)\frac{400}{(20+X_2)^2} = \frac{400X_1}{(20+X_2)^2}$$

Подставим (4):

$$a_{12} = \frac{400(1+p_5)X_2}{p_4(20+X_2)^2}$$

$$a_{21} = -p_2 p_4 E(X_2)$$

Подставим (5):

$$a_{21} = -p_4\frac{(1+p_5)X_2}{p_4 - (1+p_5)X_2}$$

$$a_{22} = -(1+p_5) + p_2 p_4 (1-X_1)E(X_2)\frac{400}{(20+X_2)^2} =$$

$$= -(1+p_5) + \frac{400 p_4 X_1}{(20+X_2)^2}$$

Подставим (4):

$$a_{22} = (-1+p_5) + \frac{400(1+p_5)X_2}{(20+X_2)^2} = (1+p_5)\left(-1 + \frac{400X_2}{(20+X_2)^2}\right)$$

$$J = \begin{bmatrix} -\frac{p_4}{p_4 - (1+p_5)X_2} & \frac{400(1+p_5)X_2}{p_4(20+X_2)^2} \\ -\frac{p_4(1+p_5)X_2}{p_4 - (1+p_5)X_2} & -(1+p_5) + \frac{400(1+p_5)X_2}{(20+X_2)^2} \end{bmatrix}$$

2 стр.

$a_{11} + a_{22} = 0$

$$-\frac{p_4}{p_4(1+p_5)x_2} - (1+p_5) + \frac{400(1+p_5)x_2}{(20+x_2)^2} = 0$$

Пусть $y = 1 + p_5$

$$-\frac{p_4}{p_4 - yx_2} - y + \frac{400yx_2}{(20+x_2)^2} = 0 \quad / \cdot p_4 - yx_2$$

$$-p_4 - y(p_4 - yx_2) + \frac{400yx_2}{(20+x_2)^2}(p_4 - yx_2) = 0$$

$$-p_4 - p_4 y + x_2 y^2 + \frac{400p_4x_2}{(20+x_2)^2}y - \frac{400x_2^2}{(20+x_2)^2}y^2 = 0$$

$$\left(x_2 - \frac{400x_2^2}{(20+x_2)^2}\right)y^2 + \left(-p_4 + \frac{400p_4x_2}{(20+x_2)^2}\right)y - p_4 = 0$$

Рассмотрим $p_4 = 12$ $x_2 = 3$

$(20+x_2)^2 = 23^2 = 529$

$$x_2 - \frac{400x_2^2}{(20+x_2)^2} = 3 - \frac{400\cdot 9}{529} = 3 - 6{,}805 \approx -3{,}805$$

$$-p_4 + \frac{400p_4x_2}{(20+x_2)^2} = -12 + \frac{400\cdot 12\cdot 3}{529} = -12 + 27{,}221 =$$

$\approx$ ~~[illegible]~~ $15{,}221$

$-3{,}805y^2 + 15{,}221y - 12 = 0$

$3{,}805y^2 - 15{,}221y + 12 = 0$

$y_1 \approx 1{,}07992$ $\quad$ $y_2 = 2{,}92034$

$p_5 = y - 1$

$p_{5,1} = 1{,}07992 - 1 = 0{,}07992$

$p_{5,2} = 2{,}92034 - 1 = 1{,}92034$ – не входа